

## Login & Registration

| Test No | Test Description                           | Test Input                                                                           | Expected Result                          |
|---------|--------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------|
| 01      | Login with valid credentials               | Email: <a href="mailto:saman@gmail.com">saman@gmail.com</a> Password: Saman@123      | Login successful                         |
| 02      | Login with invalid credentials             | Email: samangmail.com Password: saman@123                                            | Error message displayed                  |
| 03      | Create account with valid details          | Fullname: Kaveesha Kavinda<br>EmailAddress: seller@gmail.com<br>Password: Seller@123 | Account created successfully             |
| 04      | Create account with invalid details        | Fullname: Kaveesha Kavinda<br>EmailAddress: sellergmail.com<br>Password: seller@123  | Email validation error displayed         |
| 05      | Attempt SQL Injection login                | admin' OR '1'='1                                                                     | Login blocked, invalid credentials       |
| 06      | Register new user with valid informations  | Name: Kavindya, Email: user@gmail.com, Password: User@123                            | Sign in successful message               |
| 07      | Verify registration saved in Supabase Auth | Valid registration                                                                   | User data stored in authentication table |
| 08      | User logout                                | Click Logout                                                                         | Redirect to login page                   |
| 09      | Access protected dashboard without login   | Direct URL access                                                                    | Redirect to login page                   |
| 10      | Password reset request                     | Valid email(Sathsarani123@gmail.com)                                                 | Password reset link sent                 |
| 11      | Reset password                             | New valid password(Kavindya123)                                                      | Password updated successfully            |

## Product Management

| Test No | Test Description                          | Test Input                                                                           | Expected Result                      |
|---------|-------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|
| 01      | Add new product                           | Name : Lap<br>Brand : Apple<br>Quantity : 1<br>Price : 32                            | Product added successfully           |
| 02      | Edit existing product                     | Updated product details: Name : Lap<br>Brand : Apple<br>Quantity : 10<br>Price : 320 | Product updated successfully         |
| 03      | Delete product                            | Click Delete                                                                         | Product removed from list            |
| 04      | Upload product image                      | Click on add button                                                                  | Successfully uploaded image          |
| 05      | Verify my product                         | product details like name, brand, type, price, stock                                 | Display correct details              |
| 06      | Choose a spec type to add to this product | RAM Type, Graphics, Refresh Rate, Operating system, storage, display type            | Correctly display choses information |

## Auction & bid Management

| Test No | Test Description                     | Test Input                                  | Expected Result                                                             |
|---------|--------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------|
| 01      | Create auction                       | Valid product and price                     | Auction created successfully                                                |
| 02      | Verify countdown timer               | 5-minute auction                            | Timer decreases every second                                                |
| 03      | Verify auction status update         | Start/End/Cancel action                     | Status updated correctly                                                    |
| 04      | Place valid bid on active auction    | Current price: 100000,<br>User bid: 1001001 | Bid placed successfully                                                     |
| 05      | Submit invalid bid on active auction | Current highest: 20000,<br>New bid: 1001001 | Should show error message<br>"Bid must be higher than current bid of 20000" |
| 06      | Refresh after bid placed             | Browser refresh                             | Highest bid remains visible                                                 |

|    |                                                    |                              |                                                    |
|----|----------------------------------------------------|------------------------------|----------------------------------------------------|
| 07 | Verify auction details stored in Supabase database | Valid create auction request | Auction row should be inserted into auctions table |
|----|----------------------------------------------------|------------------------------|----------------------------------------------------|

### Shopping Cart Management

| Test No | Test Description                                             | Test Input                                                 | Expected Result                                 |
|---------|--------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------|
| 01      | Add product or multiple products to cart                     | Click Add to Cart                                          | Product added successfully                      |
| 02      | Verify details transferred to order summery                  | Display total amount with subtotal and tax                 | Should Display total amount correctly           |
| 03      | Add items and inspect browser local storage                  | Add products                                               | Cart JSON data should save in local storage     |
| 04      | Remove product from cart                                     | Click Remove                                               | Product removed from cart                       |
| 05      | Enter correct shipping information and display order summery | Name, Email Address, Phone number, Shipping Address, City, | Display total amount correctly in order summery |
| 06      | Click back to cart                                           | Click on button                                            | Correctly redirect to shopping cart             |

### Order & Checkout Management

| Test No | Test Description                                                     | Test Input                            | Expected Result                             |
|---------|----------------------------------------------------------------------|---------------------------------------|---------------------------------------------|
| 01      | Placed order with all valid customer detail                          | Name, Email Address, Phone number     | Order placed successfully                   |
| 02      | Verify new order inserted into Supabase                              | Submit order request                  | New order row should be created in database |
| 03      |                                                                      |                                       |                                             |
| 04      | Verify display correct total amount with correct user data like name | Place order                           | Display correct amount, user details        |
| 05      | Verify payment status display correctly                              | Display state from like pending, paid | Display correct payment status              |

|    |                                         |                     |                             |
|----|-----------------------------------------|---------------------|-----------------------------|
| 06 | Update order status Pending → Delivered | Pending → Delivered | Status updated successfully |
|----|-----------------------------------------|---------------------|-----------------------------|

#### Admin Dashboard & Reporting

| Test No | Test Description                                        | Test Input            | Expected Result                                |
|---------|---------------------------------------------------------|-----------------------|------------------------------------------------|
| 01      | Verify total revenue amount shown from completed orders | Delivered orders      | Correct revenue displayed                      |
| 02      | Verify number of live auctions displayed                | Active auctions in DB | Correct active auction count should show       |
| 03      | Verify registered users count display                   | Users table records   | Correct user count displayed                   |
| 04      | Update user role                                        | Admin role request    | Role updated successfully                      |
| 05      | Verify pending order quantity shown                     | Pending orders exist  | Correct pending order count should display     |
| 06      | Verify Supabase count/sum queries execute correctly     | Fetch dashboard stats | Correct aggregate values should return from DB |

#### System Integration & Data Validation

| Test No | Test Description                      | Test Input             | Expected Result                       |
|---------|---------------------------------------|------------------------|---------------------------------------|
| 01      | Validate client-server communication  | Valid mutation request | Data transferred and stored correctly |
| 02      | Render Auctions Client data correctly | Auction dataset        | Auction data displayed correctly      |
| 03      | Render Orders Clients data correctly  | Order dataset          | Order data displayed correctly        |
| 04      | Verify state updates correctly        | State change action    | State updated correctly               |

|    |                                      |                     |                                     |
|----|--------------------------------------|---------------------|-------------------------------------|
| 05 | Verify server-client data separation | Server fetched data | Data rendered client-side correctly |
| 06 | Verify dashboard page performance    | Large dataset       | Page loads efficiently              |

### Search & Filter System

| Test No | Test Description                      | Test Input                                                     | Expected Result                                       |
|---------|---------------------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| 01      | Apply single filters                  | One filter parameters like product type                        | Correct filtered results displayed                    |
| 02      | Apply multiple filters simultaneously | Multiple filter parameters like product type, brand, processor | Correct filtered results displayed                    |
| 03      | Validate order filtering logic        | Order filter conditions                                        | Correct filtering logic executed                      |
| 04      | Search with valid keyword             | Search Product name like apple                                 | Matching records returned                             |
| 05      | Search with no matching results       | Search Invalid keyword                                         | No matching records returned                          |
| 06      | Verify sort by option                 | Click on dropdown                                              | Display relevance, newest arrival, price :low to high |

### Security & access control module

| Test No | Test Description                                      | Test input               | Expected result                                                    |
|---------|-------------------------------------------------------|--------------------------|--------------------------------------------------------------------|
| 01      | Access protected dashboard without login              | Direct URL access        | Redirect to login page                                             |
| 02      | User attempts product delete action                   | User clicks delete       | System should block with error message like "a server error occur" |
| 03      | Unauthorized user attempts direct restricted DB query | Restricted query request | Supabase RLS should deny data access                               |
| 04      | Password reset request                                | Valid email              | Password reset link sent                                           |

|    |                                         |                                |                                                 |
|----|-----------------------------------------|--------------------------------|-------------------------------------------------|
| 05 | Reset password                          | New valid password             | Password updated successfully                   |
| 06 | Normal user tries accessing /admin page | Buyer or seller account access | Access should be denied with error msg like 404 |